



Mark Scheme (Results)

October 2022

Pearson Edexcel International Advanced
Subsidiary Level
In Chemistry (WCH12)
Paper 01: Energetics, Group Chemistry,
Halogenoalkanes and Alcohols

Section A

Question Number	Answer	Mark
1	The only correct answer is B (463.5) <i>A is not correct because 242 has been subtracted not added</i> <i>C is not correct because 498 has not been divided by 2</i> <i>D is not correct because the final answer has not been divided by 2</i>	1

Question Number	Answer	Mark
2	The only correct answer is A ($\text{Na(s)} + \frac{1}{2}\text{Cl}_2\text{(g)} \rightarrow \text{NaCl(s)}$) <i>B is not correct because the enthalpy change of formation refers to only one mole of a compound</i> <i>C is not correct because the Na should be solid and the Cl should be $\frac{1}{2}\text{Cl}_2$</i> <i>D is not correct because ions are not involved in the enthalpy change of formation</i>	1

Question Number	Answer	Mark
3	The only correct answer is D (-193 kJ mol^{-1}) <i>A is not correct because $4 \times H_2$ has been used, not 2 and the sign is incorrect</i> <i>B is not correct because $4 \times H_2$ has been used, not 2</i> <i>C is not correct because the sign is incorrect</i>	1

Question Number	Answer	Mark
4	The only correct answer is C (0.72 g) <i>A is not correct because a 2:1 ratio has been used, not 1:2</i> <i>B is not correct because a 1:1 ratio has been used, not 1:2</i> <i>D is not correct because a 1:3 ratio has been used, not 1:2</i>	1

Question Number	Answer	Mark
5	The only correct answer is C (solubility of the sulfates) <i>A is not correct because the reactivity of the elements increases down the group</i> <i>B is not correct because the solubility of the hydroxides increases down the group</i> <i>D is not correct because the thermal stability of the carbonates increases down the group</i>	1

Question Number	Answer	Mark
6	The only correct answer is C (4) <i>A is not correct because there are 4 isomers</i> <i>B is not correct because there are 4 isomers</i> <i>D is not correct because there are 4 isomers</i>	1

Question Number	Answer	Mark
7	The only correct answer is D (nucleophilic substitution) <i>A is not correct because the CN^- ion is a nucleophile and the reaction is a substitution</i> <i>B is not correct because the reaction is a substitution</i> <i>C is not correct because the CN^- ion is a nucleophile</i>	1

Question Number	Answer	Mark
8	The only correct answer is D (energy emitted, from excited state to ground state) <i>A is not correct because the flame colour is not caused by the absorption of energy</i> <i>B is not correct because electrons do not emit energy on promotion</i> <i>C is not correct because electrons do not absorb energy on returning to the ground state</i>	1

Question Number	Answer	Mark
9	The only correct answer is A (hydrogen iodide has stronger London forces than hydrogen bromide) <i>B is not correct because hydrogen iodide has a smaller permanent dipole than hydrogen bromide</i> <i>C is not correct because neither HBr or HI can form hydrogen bonds</i> <i>D is not correct because the H–I bond is weaker than the H–Br bond</i>	1

Question Number	Answer	Mark
10	The only correct answer is B (NaNO_3) <i>A is not correct because both oxygen and nitrogen dioxide would be produced</i> <i>C is not correct because both oxygen and nitrogen dioxide would be produced</i> <i>D is not correct because both oxygen and nitrogen dioxide would be produced</i>	1

Question Number	Answer	Mark
11	The only correct answer is B () <i>A is not correct because propan-2-ol has a lower boiling temperature as the alcohol is branched</i> <i>C is not correct because butane has a lower boiling temperature as it does not form hydrogen bonds</i> <i>D is not correct because 2-methylpropane has a lower boiling temperature as it does not form hydrogen bonds</i>	1

Question Number	Answer	Mark
12	The only correct answer is A (HCl) <i>B is not correct because hydrogen bonds form between molecules of H_2O</i> <i>C is not correct because hydrogen bonds form between molecules of HF</i> <i>D is not correct because hydrogen bonds form between molecules of NH_3</i>	1

Question Number	Answer	Mark
13	The only correct answer is C (+5) <i>A is not correct because the oxidation number is +5 in BrO_3^-</i> <i>B is not correct because the oxidation number is +5 in BrO_3^-</i> <i>D is not correct because the oxidation number is +5 in BrO_3^-</i>	1

Question Number	Answer	Mark
14	The only correct answer is D $\text{Zn(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{ZnSO}_4\text{(aq)} + \text{Cu(s)}$ <i>A is not correct because Cl has undergone disproportionation</i> <i>B is not correct because O has undergone disproportionation</i> <i>C is not correct because Cl has undergone disproportionation</i>	1

Question Number	Answer	Mark
15	The only correct answer is D (reactivity of the elements increases) <i>A is not correct because atomic radius increases down the group</i> <i>B is not correct because the boiling temperature increases down the group</i> <i>C is not correct because electronegativity decreases down the group</i>	1

Question Number	Answer	Mark
16	The only correct answer is A (0.75) <i>B is not correct because this is the reciprocal of the correct answer</i> <i>C is not correct because this is the total volume at 40 seconds divided by 40</i> <i>D is not correct because this is the initial rate</i>	1

Question Number	Answer	Mark
17(a)	The only correct answer is C (the area under the curve to the right of the activation energy, E_a represents the number of particles with enough energy to react) <i>A is not correct because this is the mode energy of the particles</i> <i>B is not correct because the activation energy is the minimum energy required for a reaction to take place</i> <i>D is not correct because a catalyst would move the activation energy to the left</i>	1

Question Number	Answer	Mark
17(b)	The only correct answer is D (shifts to the left, higher) <i>A is not correct because the curve would shift to the left and the peak would be higher</i> <i>B is not correct because the curve would shift to the left</i> <i>C is not correct because the peak would be higher</i>	1

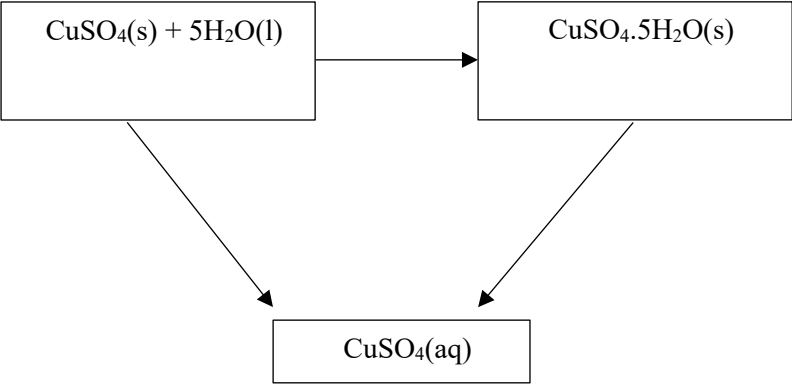
Question Number	Answer	Mark
18	The only correct answer is C (arrow 3) <i>A is not correct because the arrow 1 is correctly used</i> <i>B is not correct because the arrow 2 is correctly used</i> <i>D is not correct because the arrow 4 is correctly used</i>	1

Question Number	Answer	Mark
19	The only correct answer is B (yellow to orange and pink to colourless) <i>A is not correct because the phenolphthalein colour change is reversed and the methyl orange colour change is for acid to neutral</i> <i>C is not correct because the methyl orange colour change is for acid to neutral</i> <i>D is not correct because the phenolphthalein colour change is reversed</i>	1

Total for Section A = 20 Marks

Section B

Question Number	Answer	Additional Guidance	Mark
20(a)	- M1 moles of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (1) - M2 energy change (1) - M3 enthalpy change per mole (1) - M4 correct sign and units and 2 or 3 SF (1) Note M4 is not a stand-alone mark it depends on a sensible calculation by dividing joules by a number of moles.	<u>Example of calculation:</u> $10.68 \div 249.6 = 0.042788 / 89 \div 2080$ $2.5 \times 55 \times 3.70 = 508.75 \text{ (J)} = 0.50875 \text{ (kJ)}$ $508.75 \div 0.042788 = 11890 \text{ (J mol}^{-1}\text{)}$ Or $0.50875 \div 0.042788 = 11.890 \text{ (kJ mol}^{-1}\text{)}$ $(+)11900 \text{ J mol}^{-1} / (+)11.9 \text{ kJ mol}^{-1} / (+)12000 \text{ J mol}^{-1} / (+)12 \text{ kJ mol}^{-1}$ TE throughout Correct answer with sign and units and 2-3 SF score (4)	4

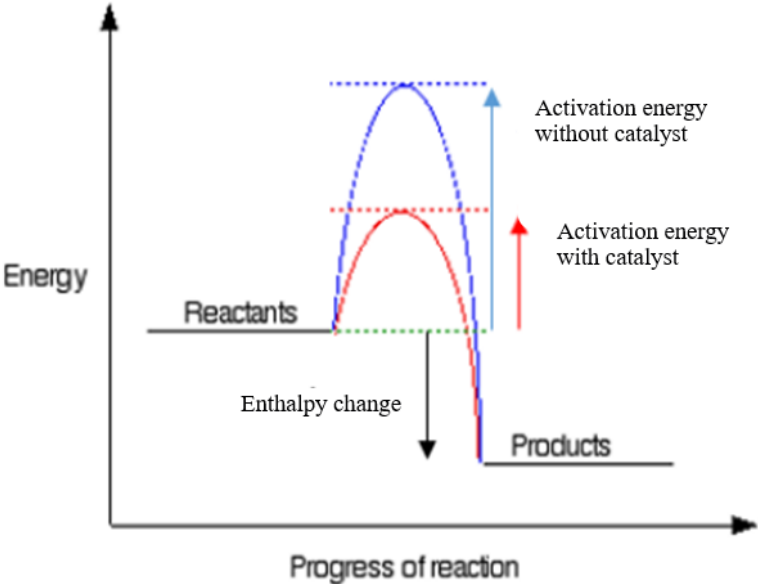
Question Number	Answer	Additional Guidance	Mark
20(b)(i)	A diagram which shows • both arrows pointing down • correct product in box	Example of diagram  <pre> graph TD A["CuSO4(s) + 5H2O(l)"] --> B["CuSO4.5H2O(s)"] A --> C["CuSO4(aq)"] B --> C </pre> Ignore any extra water in the box e.g. + 5H₂O Ignore any numbers on the arrows Allow ions separated Cu²⁺ (aq) and SO₄²⁻ (aq) Allow CuSO₄ + (aq)	2

Question Number	Answer	Additional Guidance	Mark
20(b)(ii)	- correct use of data - correct sign and answer Note the only TE is using their value from (a).	<u>Example of calculation:</u> (+) $-67.4 \text{ (kJ mol}^{-1}\text{)}$ (-) $+11.9 \text{ (kJ mol}^{-1}\text{)}$ $-79.3 / -79 \text{ (kJ mol}^{-1}\text{)}$ Ignore units unless wrong but not award mixed units Allow $-79.16 \text{ (kJ mol}^{-1}\text{)}$ for rounding moles to 0.43 Allow TE on value from (a). No TE on incorrect cycle Ignore SF	2

Total for Question 20 = 8 Marks

Question Number	Answer	Additional Guidance	Mark
21(a)(i)	An answer that makes reference to the following points: • both have London forces (only) (1) • S molecules have more electrons (1) / S is a larger molecule (than oxygen) / S electrons are more easily polarised Independent marks	Accept dispersion forces Accept instantaneous dipole-induced dipoles Allow van der Waals' forces Note any mention of other intermolecular forces being present e.g. dipole-dipole negates M1 Allow S₈, rather than O₂ Allow just S has more electrons Allow just S is larger Allow reverse argument Ignore electron density Do not award a S atom has more electrons than an oxygen molecule	

Question Number	Answer	Additional Guidance	Mark
21(a)(ii)	An answer that makes reference to the following points: Increasing temperature - equilibrium shifts to the LHS/backwards (1) - in the endothermic direction (in order to reduce the temperature) (1) Increasing pressure - equilibrium shifts to the RHS/forward (1) - to the side with fewer (gaseous) molecules/moles (in order to reduce the pressure) (1) Independent marks	Ignore any reference to yield Allow reaction is exothermic Allow favours endothermic direction Allow favours RHS/forward direction Allow 3 moles (of gas) on the LHS (forms) 2 moles(of gas) on the RHS. If numbers of moles are quoted they must be correct. Allow reverse argument Ignore any reference to rate	4

Question Number	Answer	Additional Guidance	Mark
21(a)(iii)	- reactants/(2)SO₂ + O₂ higher than products/(2)SO₃ (1) - non-catalysed activation energy correctly labelled and arrow going up (1) - catalysed activation energy correctly labelled and arrow going up (1) - enthalpy change labelled and arrow going down (1) If just one curve is drawn max (3) MB distribution scores (0) If endothermic reaction TE available for M2, M3 and M4 Double-headed arrows or no arrow heads penalise once	 Allow energy for enthalpy Allow E_a and E_{cat} Allow intermediates in the activation energy curves Ignore sign and units of energy even if incorrect	4

Question Number	Answer	Additional Guidance	Mark
21(a)(iv)	An explanation that makes reference to the following points: • (operates at) a lower temperature (1) • less energy required (for the reaction to proceed at an acceptable rate) or less burning of fossil fuels / less greenhouse gas evolved / less CO₂ evolved (1)	Allow lower temperature Allow less energy costs Ignore just cheaper Ignore less time/faster rate Allow less pollution	2

Question Number	Answer	Additional Guidance	Mark																				
*21(b)	This question assesses the student’s ability to show a coherent and logically structured answer with linkages and fully sustained reasoning Marks are awarded for indicative content and for how the answer is structured and shows lines of reasoning. The following table shows how the marks should be awarded for indicative content. <table><tr><th>Number of indicative marking points seen in answer</th><th>Number of marks awarded for indicative marking points</th></tr><tr><td>6</td><td>4</td></tr><tr><td>5-4</td><td>3</td></tr><tr><td>3-2</td><td>2</td></tr><tr><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td></tr></table> The following table shows how the marks should be awarded for structure and lines of reasoning <table><tr><th></th><th>Number of marks awarded for structure of answer and sustained lines of reasoning</th></tr><tr><td>Answer shows a coherent logical structure with linkages and fully sustained lines of reasoning demonstrated throughout</td><td>2</td></tr><tr><td>Answer is partially structured with some linkages and lines of reasoning</td><td>1</td></tr><tr><td>Answer has no linkages between points and is unstructured</td><td>0</td></tr></table>	Number of indicative marking points seen in answer	Number of marks awarded for indicative marking points	6	4	5-4	3	3-2	2	1	1	0	0		Number of marks awarded for structure of answer and sustained lines of reasoning	Answer shows a coherent logical structure with linkages and fully sustained lines of reasoning demonstrated throughout	2	Answer is partially structured with some linkages and lines of reasoning	1	Answer has no linkages between points and is unstructured	0	Guidance on how the mark scheme should be applied. The mark for indicative content should be added to the mark for lines of reasoning. For example, a response with five indicative marking points that is partially structured with some linkages and lines of reasoning scores 4 marks (3 marks for indicative content and 1 mark for partial structure and some linkages and lines of reasoning). If there were no linkages between the points, then the same indicative marking points would yield an overall score of 3 marks (3 marks for indicative content and no marks for linkages). In general it would be expected that 5 or 6 indicative points would get 2 reasoning marks, and 3 or 4 indicative points would get 1 mark for reasoning and 0, 1 or 2 indicative points would score zero marks for reasoning. If there is any incorrect chemistry, deduct mark(s) from the reasoning. If no reasoning mark(s) awarded do not deduct mark(s). Comment: Look for the indicative marking points first, then consider the mark for the structure of the answer and sustained line of reasoning	6
Number of indicative marking points seen in answer	Number of marks awarded for indicative marking points																						
6	4																						
5-4	3																						
3-2	2																						
1	1																						
0	0																						
	Number of marks awarded for structure of answer and sustained lines of reasoning																						
Answer shows a coherent logical structure with linkages and fully sustained lines of reasoning demonstrated throughout	2																						
Answer is partially structured with some linkages and lines of reasoning	1																						
Answer has no linkages between points and is unstructured	0																						

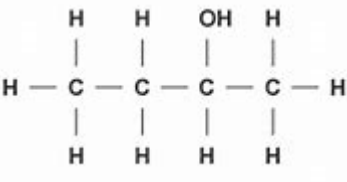
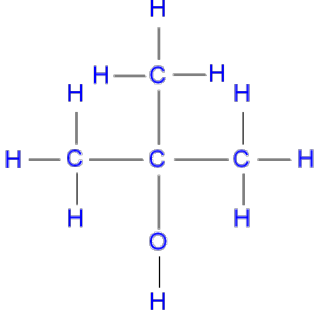
	Indicative content <u>Potassium chloride</u> IP1 (misty/steamy fumes of) HCl/hydrogen chloride <u>Potassium iodide</u> IP2 purple vapour/black solid and I₂/iodine IP3 bad eggs and H₂S/ hydrogen sulfide OR yellow solid and S/ sulfur OR Choking gas and SO₂/sulfur dioxide IP4 KCl is not a redox reaction and KI is a redox reaction / KCl cannot reduce sulfuric acid but KI can IP5 iodide stronger reducing agent than chloride	Note observation is not required for IP This can be seen in an equation Ignore any reference to hydrogen iodide/HI/misty fumes The two parts of this answer may often be seen in different places. Allow hydrogen iodide or potassium iodide is a stronger reducing agent than hydrogen chloride or potassium chloride. Allow reducing ability (of the halides) increases down the group Or reverse argument Do not award iodine/I is a stronger reducing agent than chlorine/Cl	
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IP6 (because) S is reduced from +6 to -2 in H_2S OR S is reduced from +6 to 0 in S OR S is reduced from +6 to +4 in SO_2 OR Any balanced equation making H_2S, SO_2, or S showing electrons eg $8\text{H}^+ + \text{H}_2\text{SO}_4 + 8\text{e}^- \longrightarrow \text{H}_2\text{S} + 4\text{H}_2\text{O}$ $2\text{H}^+ + \text{H}_2\text{SO}_4 + 2\text{e}^- \longrightarrow \text{SO}_2 + 2\text{H}_2\text{O}$ $6\text{H}^+ + \text{H}_2\text{SO}_4 + 6\text{e}^- \longrightarrow \text{S} + 4\text{H}_2\text{O}$	Allow just correct stated product and oxidation number eg -2 in H_2S or 0 in S or $+4$ in SO_2 Ignore I_2 oxidation number = 0
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Question Number	Answer	Additional Guidance	Mark
21(c)	- number of moles of sulfuric acid in 40.5 cm³ (1) - concentration of the concentrated sulfuric acid (1)	<u>Example of calculation</u> $1.5 \times 500 \div 1000 = 0.75 \text{ (mol)}$ $0.75 \times 1000 \div 40.5 = 18.519 \text{ (mol dm}^{-3}\text{)}$ Allow TE on incorrect moles Units not required but if given they must be correct Correct answer with or without working scores (2) Ignore SF except 1 SF	2

Total for Question 21 = 20 marks

Question Number	Answer	Additional Guidance	Mark
22(a)	An explanation that makes reference to the following points: - contains the OH group/OH is the alcohol group (1) - primary means the C bonded to the OH (group) is attached to 1 (or 0) alkyl group / carbon atom/ R group (1)	Accept a hydrocarbon in which one H atom has been replaced by an OH group Allow contains O-H bond Allow contains a C-OH bond Ignore hydroxyl Do not award hydroxide/ OH⁻ Allow the OH group is attached to a carbon atom bonded to 2 (or 3) H atoms Do not award the OH is attached to a single R/alkyl/carbon group	2

Question Number	Answer				Additional Guidance	Mark
22(b)	Name	butan-1-ol		(2-)methylpropan-2-ol	(2) One mark for each name	5
	Displayed formula				(2) One mark for each correct displayed formula	
	Nature of alcohol	primary	secondary		(1) One mark for both alcohol types Allow CH ₃ and OH undisplayed Penalise missing Hs only once.	

Question Number	Answer	Additional Guidance	Mark
22(c)(i)	An answer that makes reference to the following points: - potassium dichromate(VI) / $\text{K}_2\text{Cr}_2\text{O}_7$ (1) - sulfuric acid / H_2SO_4 (1) Note M2 depends on M1 or a near miss such as potassium permanganate or potassium dichromate with the wrong oxidation number	Allow sodium dichromate(VI) / $\text{Na}_2\text{Cr}_2\text{O}_7$ If oxidation state is given it must be correct Do not award potassium manganate(VII) / potassium permanganate Do not award hydrochloric acid Ignore just 'acidified' Ignore concentration of sulfuric acid Ignore heat If no other mark is scored acidified dichromate / H^+ and $\text{Cr}_2\text{O}_7^{2-}$ score (1)	2

Question Number	Answer	Additional Guidance	Mark
22(c)(ii)	$(\text{CH}_3)_2\text{CHCH}_2\text{OH} + [\text{O}] \rightarrow (\text{CH}_3)_2\text{CHCHO} + \text{H}_2\text{O}$	Allow $\text{C}_4\text{H}_{10}\text{O} + [\text{O}] \rightarrow \text{C}_4\text{H}_8\text{O} + \text{H}_2\text{O}$ The [O] may be above the arrow Allow displayed/molecular/skeletal	1

Question Number	Answer	Additional Guidance	Mark
22(c)(iii)	An answer that makes reference to two of the following points: • (resulting mixture) would give a peak due to O–H bond in alcohols at 3750-3200 (cm^{-1}) • would give a peak due to C=O bond in aldehydes at 1740-1720 (cm^{-1}) or would give a peak due to C–H bond in CHO at 2900-2820 and/ or 2775-2700 (cm^{-1})	Ignore any reference to the size/width of the peaks due to stretching etc Allow (resulting mixture) would give a peak due to OH (in alcohols) at 3750-3200 (cm^{-1}) Do not award -OH Do not award 1720-1700 (cm^{-1}) for C=O in ketones Allow any range within the range. Do not award single numbers but penalise once only If no other marks have been scored a correct wave number range for both the O-H and C=O score (1) 3750-3200 (cm^{-1}) 1740-1720 (cm^{-1})	(1) (1)

(Total for Question 22 = 12 Marks)

Section C

Question Number	Answer	Additional Guidance	Mark
23(a)(i)	$\text{TiO}_2 + 2\text{Cl}_2 + 2\text{C} \rightarrow \text{TiCl}_4 + 2\text{CO}$	Ignore state symbols even if incorrect Allow multiples	1

Question Number	Answer	Additional Guidance	Mark
23(a)(ii)	- Ti is reduced and from +4 to 0 - Mg is oxidised and from 0 to +2	(1) Four correct oxidation numbers with no or incorrect mention of reduced or oxidised scores (1) (1) Ti is reduced as it gains electrons and Mg is oxidised as it loses electrons with no or incorrect oxidation numbers scores (1)	2

Question Number	Answer	Additional Guidance	Mark
23(b)	- hydrolysis - titanium((IV)) oxide/TiO_2	(1) Do not award hydration (1) Ignore hydrogen chloride/HCl Allow titanium((IV))hydroxide/$\text{Ti}(\text{OH})_4$ If both name and formula/oxidation number are given they must all be correct	2

Question Number	Answer	Additional Guidance	Mark
23(c)	$\left[\begin{array}{cccc} \text{H} & \text{CH}_3 & \text{H} & \text{CH}_3 \\ & & & \\ -\text{C} & -\text{C} & -\text{C} & -\text{C}- \\ & & & \\ \text{H} & \text{CH}_3 & \text{H} & \text{CH}_3 \end{array} \right]_{n/2}$ • correct repeating unit • two repeating units and extension Note M2 depends on M1 or near miss eg missing a H	(1) Allow adjacent pairs of methyl groups (1) Ignore square brackets and subscript n/2 Ignore connectivity of CH₃	2

Question Number	Answer	Additional Guidance	Mark
23(d)(i)	An answer that makes reference to the following points: - reagent: potassium hydroxide / KOH (1) - conditions: aqueous/water (1) M2 dependent on M1 or near miss e.g. OH⁻ will not score M1 but will allow access to M2.	Allow sodium hydroxide / NaOH Ignore any reference to concentration or heat Allow dilute	2

Question Number	Answer	Additional Guidance	Mark
23(d)(ii)	An answer that makes reference to the following points: - reagent: (concentrated) phosphoric(V) acid / H₃PO₄ (1) - reaction type: elimination/dehydration (1) Independent marks	Allow (concentrated) sulfuric acid Ignore heat Do not award H₃PO₃ Allow passing vapour over porous pot / alumina or any surface catalyst and heat	2

Question Number	Answer	Additional Guidance	Mark
23(d)(iii)	An answer that makes reference to the following points: - reagent: phosphorus(V) chloride/ phosphorus pentachloride/PCl₅ (1) - equation: $\text{C}_4\text{H}_9\text{OH} + \text{PCl}_5 \rightarrow \text{C}_4\text{H}_9\text{Cl} + \text{HCl} + \text{POCl}_3$ (1)	Allow thionyl chloride / SOCl₂ Allow phosphorus (III) chloride / phosphorus trichloride / PCl₃ Allow $\text{C}_4\text{H}_9\text{OH} + \text{SOCl}_2 \rightarrow \text{C}_4\text{H}_9\text{Cl} + \text{HCl} + \text{SO}_2$ $3\text{C}_4\text{H}_9\text{OH} + \text{PCl}_3 \rightarrow 3\text{C}_4\text{H}_9\text{Cl} + \text{H}_3\text{PO}_3$ Allow skeletal, structural, displayed or molecular formulae	2

Question Number	Answer	Additional Guidance	Mark
23(d)(iv)	An answer that makes reference to the following points: - conditions: alcoholic (1) - role of the hydroxide ions: base (1) Note answers may be given on either line	Allow ethanol/ethanolic Ignore heat and concentration Allow proton acceptor/reacts with H⁺	2

Question Number	Answer	Additional Guidance	Mark
23(e)	An answer that makes reference to the following points: - moles of carbon dioxide (1) - conversion of cm³ to m³ (1) - conversion of °C to K (1) - rearrangement of the ideal gas equation (1) - calculation of pressure and correct units given (1)	<u>Examples of calculation</u> 16 ÷ 44 = 0.36364 (mol)/ 4÷11(mol) 20 ÷ 1000000 = 2 × 10⁻⁵ / 0.00002 (m³) 273 + 25 = 298 (K) $p = \frac{nRT}{V}$ $\frac{0.3636 \times 8.31 \times 298}{2 \times 10^{-5}} = 45025000 \text{ Pa} / 45025 \text{ kPa}$ 4.5 x 10⁷ Pa / 45MPa Ignore SF except 1 Allow TE for answers to M1, M2 and M3 But no TE on wrong rearrangement of gas equation Correct answer, including units with or without working scores (5)	5

Total for Question 23 = 20 Marks
Total for Section C = 20 Marks
TOTAL FOR PAPER = 80 MARKS